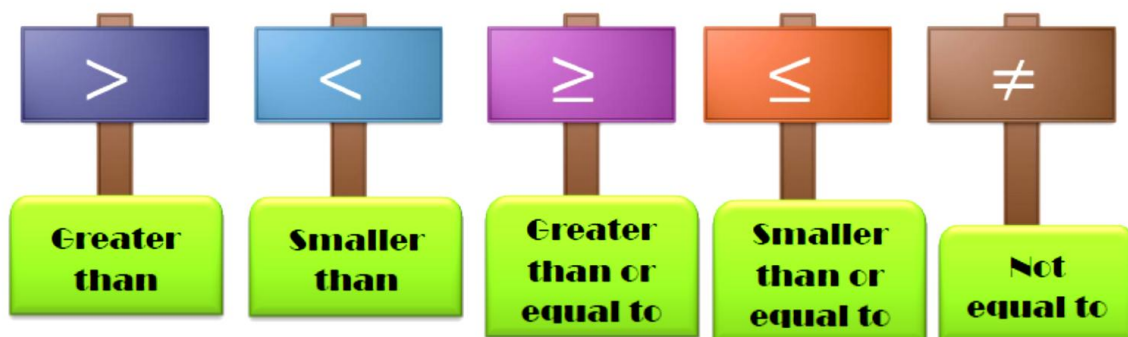


LESSON 3 : INEQUALITY



Checkpoint 1

$$\begin{aligned} 2x - 1 &= 5 \\ 2x &= 6 \\ x &= 3 \end{aligned}$$

$$\begin{aligned} 2x - 1 &> 5 \\ 2x &> 6 \\ x &> 3 \end{aligned}$$

$$\begin{aligned} x + 3 &< 5 \\ x &< 5 - 3 \\ x &< 2 \end{aligned}$$

$$\begin{aligned} x - 2 &> 10 \\ x &> 10 + 2 \\ x &> 12 \end{aligned}$$

$$\begin{aligned} 5 &< 2x - 1 \\ 6 &< 2x \\ 3 &< x \\ x &> 3 \end{aligned}$$

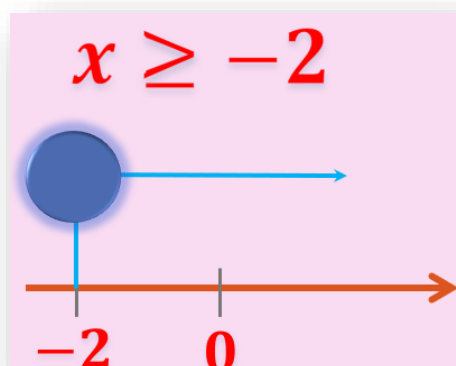
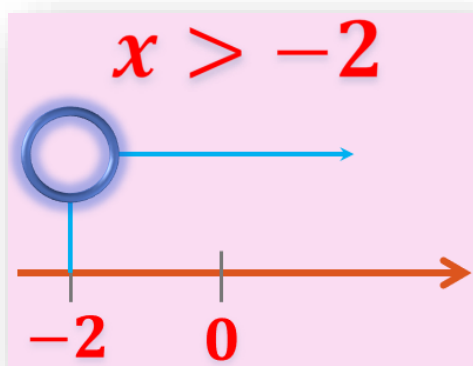


Make subject

主角放左邊

唔好冇大有細

Checkpoint 2



Let's get started!

Solve the following inequalities graphically.

1. $x + 3 < 7$

2. $x - 3 > 7$

3. $x + 6 \leq 10$

4. $7 < x - 15$

5. $11 \geq 3 + x$

6. $10 + x < 27$

7. $10 < 15 - x$

8. $209 \leq 300 - x$

9. $287 \geq 106 + x$

10. $256 - x < 96$

Checkpoint 3

$$\begin{aligned}5x &\leq 15 \\ x &\leq \frac{15}{5} \\ x &\leq 3\end{aligned}$$

$$\begin{aligned}\frac{x}{4} &\geq 2 \\ x &\geq 2 \times 4 \\ x &\geq 8\end{aligned}$$

You have learnt more!

Solve the following inequalities graphically.

11. $3x < 15$

12. $2x < 86$

13. $\frac{x}{8} < 9$

14. $\frac{x}{7} \leq 13$

15. $15 < \frac{x}{6}$

16. $32 \geq \frac{x}{3}$

17. $\frac{24x}{5} \geq 48$

18. $258 < \frac{3x}{2}$

19. $9 < \frac{3x}{35}$

20. $\frac{3x}{7} < 8$

21. $\frac{5x}{18} < 10$

22. $\frac{216x}{9} \leq 24$

Checkpoint 4

$$\begin{aligned} -\frac{x}{4} &= 2 \\ -x &= 2 \times 4 \\ -x &= 8 \\ -8 &= x \\ x &= -8 \end{aligned}$$

$$\begin{aligned} -\frac{x}{4} &\geq 2 \\ -x &\geq 2 \times 4 \\ -x &\geq 8 \\ \text{開口} \quad \text{尖頭} \\ -8 &\geq x \\ x &\leq -8 \\ \text{尖頭} \quad \text{開口} \end{aligned}$$

$$-x < 4$$

$$x > -4$$



乘除負數

大細調轉

Keep Fighting!

23. $-x + 3 > 7$

24. $-2x + 5 < 17$

25. $x + 2 \geq 4 + 3x$

26. $13 \leq 9x - 5$

27. $-4x + 7 > 5$

28. $2x + 3 < 10x - 5$

29. $10 - 2x \geq x + 1$

30. $5 - 3x \leq -1 + 3x$

31. $x + 4 > 4x - 8$

32. $-30 + 12x < 42$

33. $4(x - 1) \geq 7x - 10$

34. $6(3 - x) \leq 4(-2x + 3)$

PRE S3 MATHS L3 - INEQUALITY

35. $2x + 3 > 5(2x - 1)$

36. $6(3 - x) < 4(-2x + 3)$

37. $12 - 2x \geq x + 3$

38. $4 - 2x \leq -1 + 2x$

39. $x + 4 > 4x - 8$

40. $5x - 1 < 3x - 7$

41. $4x - 12 \geq 10 - 7x$

42. $13 - 5x \leq 23 - 7x$

43. $-6(5 - 2x) > 42$

44. $5(x - 8) < 10$

45. $\frac{1}{5}(-4 - x) \geq 8$

46. $\frac{1}{3}(4x - 1) \leq 1$

Give it a try

$$\begin{aligned}
 3(x+2) - 5(x+4) &> 0 \\
 3x + 6 - 5x - 20 &> 0 \\
 -2x - 14 &> 0 \\
 -2x &> 14 \\
 -14 &> 2x \\
 -7 &> x \\
 x &< -7
 \end{aligned}$$

$$\begin{aligned}
 2x + 1 &< \frac{4x + 1}{3} \\
 3(2x + 1) &< 4x + 1 \\
 6x + 3 &< 4x + 1 \\
 6x - 4x &< +1 - 3 \\
 2x &< -2 \\
 x &< -1
 \end{aligned}$$



一邊有分母 n ,

其他 terms 全部 $\times \frac{n}{n}$

Challenging!!

Case 1

$$\frac{2 - 3x}{4} \leq 1 - x$$

$$\frac{2 - 3x}{4} \leq (1 - x) \times \frac{4}{4}$$

$$2 - 3x \leq 4(1 - x)$$

$$2 - 3x \leq 4 - 4x$$

$$-3x + 4x \leq 4 - 2$$

$$x \leq 2$$

Case 2

$$\frac{5x}{3} - 2 < \frac{10x}{9} + 3$$

$$\frac{5x}{3} - \frac{10x}{9} < 3 + 2$$

$$\frac{5x}{3} \times \frac{9}{9} - \frac{10x}{9} \times \frac{3}{3} < 5 \times \frac{27}{27}$$

$$45x - 30x < 135$$

$$x < 9$$

$$\frac{9}{9} \times \frac{3}{3}$$

Final Boss!

1. $\frac{x+1}{2} \geq \frac{9x+5}{19}$

2. $\frac{3x+8}{2} \geq \frac{3-x}{5}$

3. $\frac{6-x}{11} - \frac{1-x}{4} < 0$

4. $\frac{5x-3}{4} - \frac{7x-25}{3} \leq 0$

5. $x + 16 > \frac{2x}{13} + 5$

6. $\frac{5x}{3} - 2 < \frac{10x}{9} + 3$

7. $4x + 16 + \frac{x}{2} < 13$

8. $7 - 4x > \frac{x+1}{3}$

PRE S3 MATHS L3 - INEQUALITY

9. $1 + \frac{x}{7} + \frac{x-3}{2} \geq 3(x+1)$

10. $5x - \frac{2x-9}{5} \leq x - 3$

11. $3 - \frac{x+1}{2} \geq 5 - \frac{x+2}{3}$

12. $\frac{1-2x}{3} - \frac{3-4x}{5} > -2$

13. $\frac{-3x+5}{2} \geq \frac{11-x}{3}$

14. $\frac{x}{6} + 1 \leq \frac{5}{6} - x$